

CAPSTONE PROJECT AI & DATA SCIENCE

CVD Risk Classification Pipeline

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Phase 1: Context & Vision

Problem Framing and Scientific Motivation

| Triage & Decision Support



Early Detection

Identify high-risk individuals accurately before severe clinical events occur, easing healthcare infrastructure burden.



Modern Paradigm

Shift from fragile rule-based expert systems to empirical machine learning models capable of mapping complex boundaries.



Clinical Triage

Enable medical professionals to prioritize preventative interventions through data-driven patient monitoring strategies.

Clinical Data Audit

Variable	Type	Range	Audit Findings
Age	Int (Days)	10,798 - 23,713	Requires conversion to years for clinical relevance.
Gender	Cat (1/2)	1 - 2	Encoded to binary (0/1) for pipeline compatibility.
Blood Pressure	Int (mmHg)	-150 - 16,000	Severe outliers identified; logging errors present.
Height/Weight	Int/Float	55 - 250 cm	Contains unphysical values requiring strict filtering.

| Outlier Treatment Logic

IQR Method

Flagged 1,439 records. Effective for local spread but sensitive to extreme logging anomalies in clinical vitals.

Z-Score Method

Flagged 983 records. Distorted by extreme values (e.g., 16,000 mmHg), artificially inflating standard deviation.

Strategy: Applied strict physical biomedical thresholds to retain clean sample of 63,942 patients.

Phase 2: Data Intelligence

Exploratory Analysis and Feature Strategy

| Pulse Pressure Superiority

Superior Decision Boundary

Initial expectations favored raw Systolic Blood Pressure as ultimate predictor. Visual analysis revealed surprising insight.

Engineered feature **Pulse Pressure ($ap_{hi} - ap_{lo}$)** creates significantly clearer separation between healthy and high-risk patients, particularly for demographics under age 55.

| Feature Engineering Strategy



BMI Metric

Standard clinical obesity indicator. Improved recall specifically on younger patient demographics.



Pulse Pressure

Arterial stiffness indicator. Ranked as top 3 predictor via Mutual Information analysis.



Age-Risk Score

Captures compounding risk of aging with high cholesterol. Enhanced tree model precision.

| Metric Optimization

F1

Primary Performance Target

Balancing Clinical Cost

Raw accuracy proves misleading in healthcare. Focus remains on minimizing:

- ✗ **False Negatives:** Fatal risk of undiagnosed CVD.
- ! **False Positives:** Resource waste on unnecessary clinical stress tests.

| Algorithm Benchmarking

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Tuned XGBoost	73.57%	75.39%	70.21%	72.71%	80.12%
Logistic Regression	72.84%	74.92%	68.61%	71.63%	79.28%
Random Forest	72.10%	71.21%	69.15%	70.59%	77.40%
KNN	69.80%	70.35%	68.10%	69.21%	74.50%

| Ethics & Implementation



Operational Framework

Designed strictly as **Decision-Support Tool** for clinic triage.

Operational limits include:

- No validation for pediatric patients (<18 years).
- Physiological boundary constraints for inputs.
- **Human-in-the-Loop:** Clinicians must oversee automated recommendations.



Strategic Recommendations

Integrate pipeline into early-stage triage workflows to optimize monitoring.

Actively monitor for gender-related diagnostic disparities to ensure unbiased healthcare delivery.